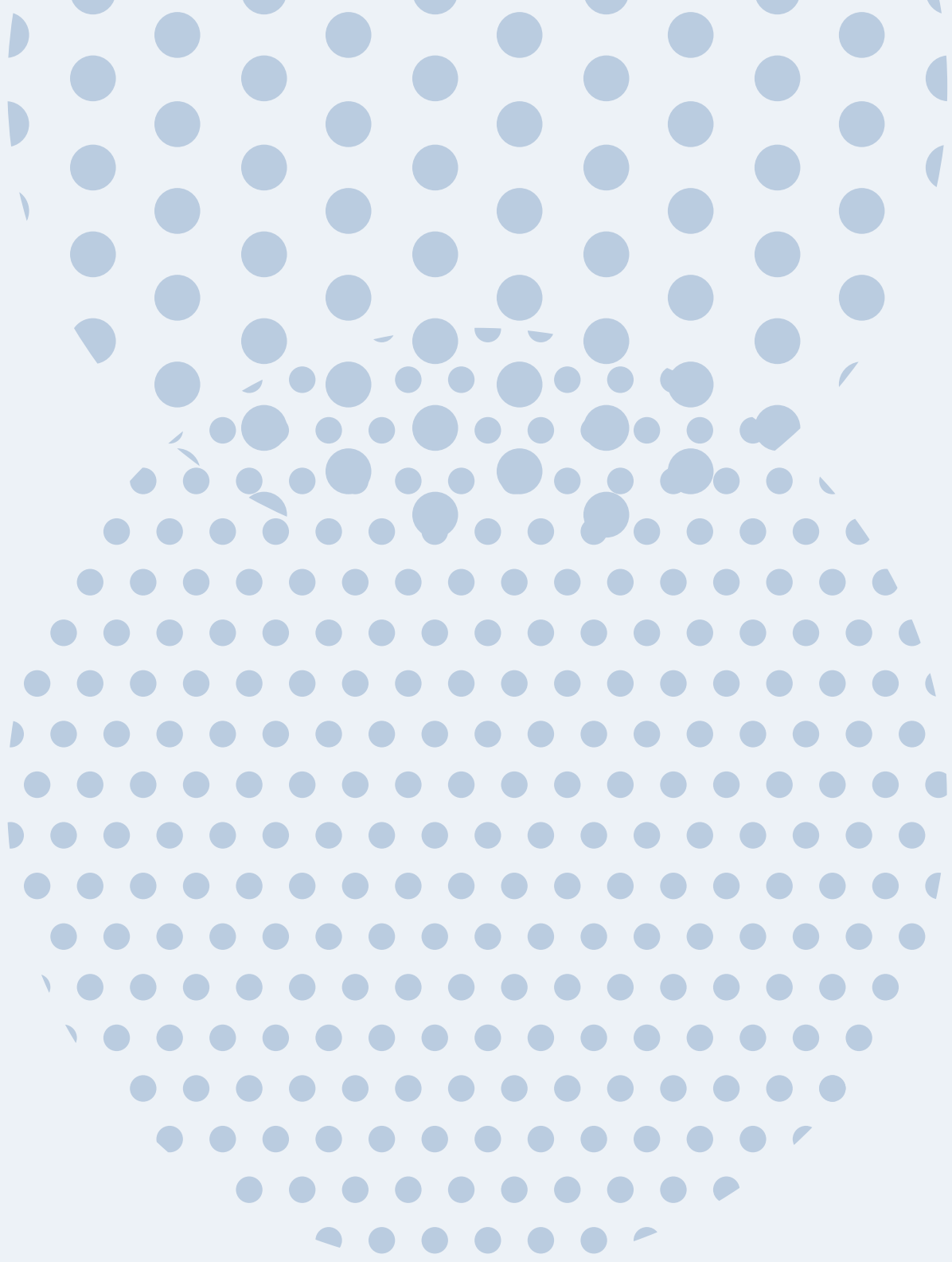




31.9 •• A 0.200 H inductor is connected in series with a $81.5\ \Omega$ resistor and an ac source. The voltage across the inductor is $v_L = -(13.0\text{ V})\sin[(474\text{ rad/s})t]$. (a) Derive an expression for the voltage v_R across the resistor. (b) What is v_R at $t = 1.93\text{ ms}$?

31.10 ●● A $250\ \Omega$ resistor is connected in series with a $4.80\ \mu\text{F}$ capacitor and an ac source. The voltage across the capacitor is $v_C = (7.60\ \text{V})\sin[(120\ \text{rad/s})t]$. (a) Determine the capacitive reactance of the capacitor. (b) Derive an expression for the voltage v_R across the resistor.

31.11 ●● A $150\ \Omega$ resistor is connected in series with a $0.250\ \text{H}$ inductor and an ac source. The voltage across the resistor is $v_R = (3.80\ \text{V})\cos[(720\ \text{rad/s})t]$. (a) Derive an expression for the circuit current. (b) Determine the inductive reactance of the inductor. (c) Derive an expression for the voltage v_L across the inductor.

31.38 ●● In an L - R - C series circuit the phase angle is 53.0° and the source voltage lags the current. The resistance of the resistor is $300\ \Omega$ and the reactance of the capacitor is $500\ \Omega$. The average power delivered by the source is $80.0\ \text{W}$. (a) What is the reactance of the inductor? (b) What is the current amplitude in the circuit? (c) What is the voltage amplitude of the source?

31.39 ●● An L - R - C series circuit has $C = 5.00\ \mu\text{F}$, $L = 0.510\ \text{H}$, and source voltage amplitude $V = 57.5\ \text{V}$. The source is operated at the resonance frequency of the circuit. If the voltage across the capacitor has amplitude $77.0\ \text{V}$, what is the value of R for the resistor in the circuit?

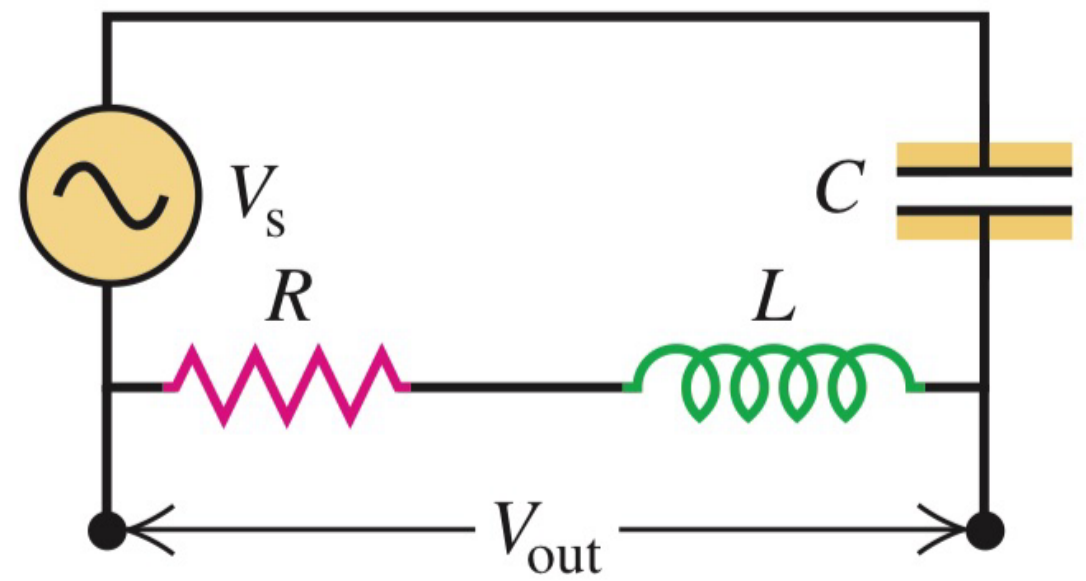
31.43 ●● A series circuit has an impedance of $58.0\ \Omega$ and a power factor of 0.720 at $45.0\ \text{Hz}$. The source voltage lags the current. (a) What circuit element, an inductor or a capacitor, should be placed in series with the circuit to raise its power factor? (b) What size element will raise the power factor to unity?

31.45 ●● In an L-R-C series circuit, $R = 380\ \Omega$, $X_C = 360\ \Omega$, and $X_L = 560\ \Omega$. The average electrical power consumed in the resistor is 60.0 W. (a) What is the power factor of the circuit? (b) What is the rms voltage of the source?

31.46 • At a frequency ω_1 the reactance of a certain capacitor equals that of a certain inductor. (a) If the frequency is changed to $\omega_2 = 5\omega_1$, what is the ratio of the reactance of the inductor to that of the capacitor? Which reactance is larger? (b) If the frequency is changed to $\omega_3 = \omega_1/7$, what is the ratio of the reactance of the inductor to that of the capacitor? Which reactance is larger? (c) If the capacitor and inductor are placed in series with a resistor of resistance R to form an L - R - C series circuit, what will be the resonance angular frequency of the circuit?

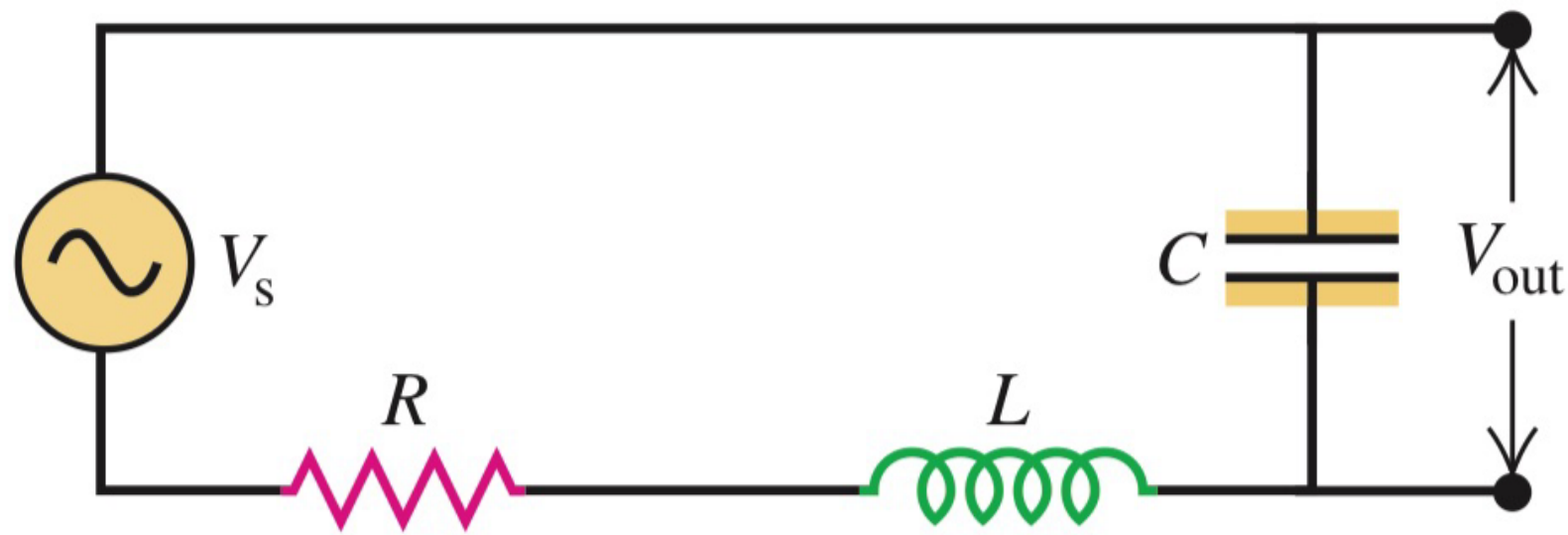
31.47 •• A High-Pass Filter. One application of L - R - C series circuits is to high-pass or low-pass filters, which filter out either the low- or high-frequency components of a signal. A high-pass filter is shown in **Fig. P31.47**, where the output voltage is taken across the L - R combination. (The L - R combination represents an inductive coil that also has resistance due to the large length of wire in the coil.) Derive an expression for V_{out}/V_s , the ratio of the output and source voltage amplitudes, as a function of the angular frequency ω of the source. Show that when ω is small, this ratio is proportional to ω and thus is small, and show that the ratio approaches unity in the limit of large frequency.

Figure **P31.47**



31.48 •• A Low-Pass Filter. Figure P31.48 shows a low-pass filter (see Problem 31.47); the output voltage is taken across the capacitor in an L - R - C series circuit. Derive an expression for V_{out}/V_s , the ratio of the output and source voltage amplitudes, as a function of the angular frequency ω of the source. Show that when ω is large, this ratio is proportional to ω^{-2} and thus is very small, and show that the ratio approaches unity in the limit of small frequency.

Figure P31.48

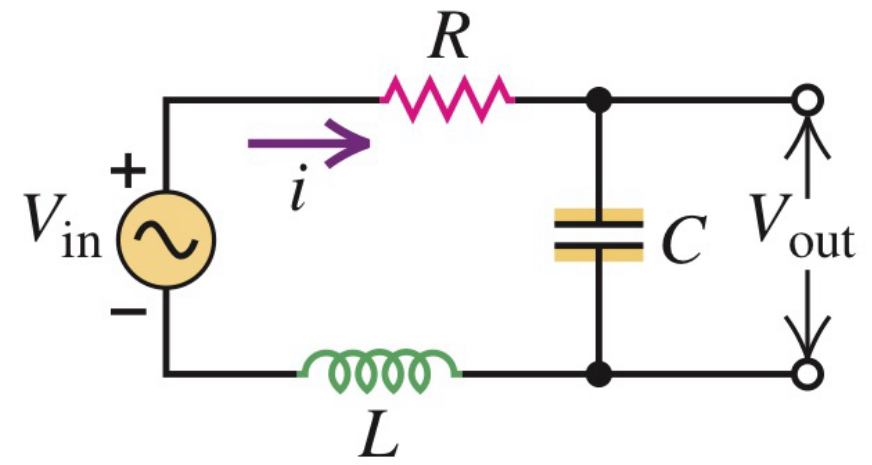


31.54 •• The L - R - C Parallel Circuit. A resistor, an inductor, and a capacitor are connected in parallel to an ac source with voltage amplitude V and angular frequency ω . Let the source voltage be given by $v = V\cos\omega t$. (a) Show that each of the instantaneous voltages v_R , v_L , and v_C at any instant is equal to v and that $i = i_R + i_L + i_C$, where i is the current through the source and i_R , i_L , and i_C are the currents through the resistor, inductor, and capacitor, respectively. (b) What are the phases of i_R , i_L , and i_C with respect to v ? Use current phasors to represent i , i_R , i_L , and i_C . In a phasor diagram, show the phases of these four currents with respect to v . (c) Use the phasor diagram of part (b) to show that the current amplitude I for the current i through the source is $I = \sqrt{I_R^2 + (I_C - I_L)^2}$. (d) Show that the result of part (c) can be written as $I = V/Z$, with $1/Z = \sqrt{(1/R^2) + [\omega C - (1/\omega L)]^2}$.

31.57 ●●● An L - R - C series circuit consists of a $2.60\ \mu\text{F}$ capacitor, a $6.00\ \text{mH}$ inductor, and a $90.0\ \Omega$ resistor connected across an ac source of voltage amplitude $15.0\ \text{V}$ having variable frequency. (a) Under what circumstances is the average power delivered to the circuit equal to $\frac{1}{2}V_{\text{rms}}I_{\text{rms}}$? (b) Under the conditions of part (a), what is the average power delivered to each circuit element and what is the maximum current through the capacitor?

31.66 ••• CALC The L - R - C series circuit shown in **Fig. P31.66** has an ac voltage source $v_{\text{in}}(t) = V_{\text{in}} \cos(\omega t)$, where V_{in} is the input voltage amplitude and ω is its angular frequency. The values of the resistance, inductance, and capacitance of the components are R , L , and C . The capacitance is variable. The output voltage is $v_{\text{out}}(t) = V_{\text{out}} \cos(\omega t + \theta)$. (a) What is the output voltage amplitude V_{out} ? (b) What is the phase angle θ of the output voltage? (c) What is the value of θ at resonance? (d) If $R = 100 \, \Omega$, $L = 1.00 \, \mu\text{H}$, and $V_{\text{in}} = 10.0 \, \text{V}$, what value of C would result in a resonance frequency of $100 \, \text{kHz}$? (e) In that case what would be the output voltage amplitude at resonance?

Figure **P31.66**



31.68 ●●● **CALC** (a) At what angular frequency is the voltage amplitude across the *resistor* in an L - R - C series circuit at maximum value? (b) At what angular frequency is the voltage amplitude across the *inductor* at maximum value? (c) At what angular frequency is the voltage amplitude across the *capacitor* at maximum value? (You may want to refer to the results of Problem 31.49.)

47. In the circuit of Figure P32.47, the battery emf is 50.0 V, the resistance is $250\ \Omega$, and the capacitance is $0.500\ \mu\text{F}$. The switch S is closed for a long time

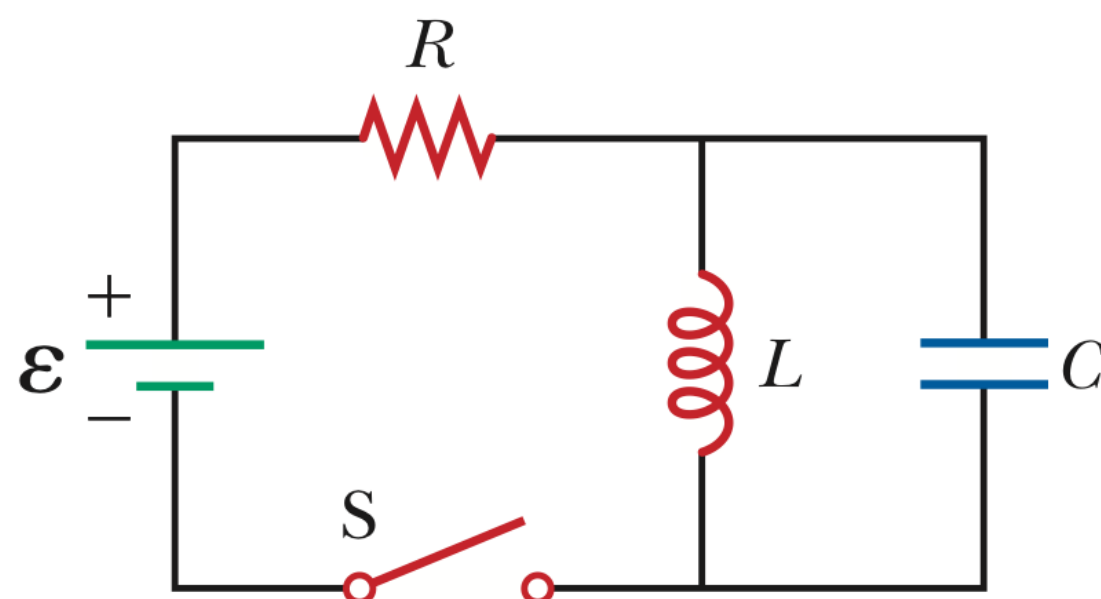


Figure P32.47

interval, and zero potential difference is measured across the capacitor. After the switch is opened, the potential difference across the capacitor reaches a maximum value of 150 V. What is the value of the inductance?

48. A $1.05\text{-}\mu\text{H}$ inductor is connected in series with a variable capacitor in the tuning section of a shortwave radio set. What capacitance tunes the circuit to the signal from a transmitter broadcasting at 6.30 MHz ?

49. A $1.00\text{-}\mu\text{F}$ capacitor is charged by a 40.0-V power supply. The fully charged capacitor is then discharged through a 10.0-mH inductor. Find the maximum current in the resulting oscillations.

50. Calculate the inductance of an LC circuit that oscillates at 120 Hz when the capacitance is $8.00\ \mu\text{F}$.

51. An LC circuit consists of a 20.0-mH inductor and a 0.500- μF capacitor. If the maximum instantaneous current is 0.100 A, what is the greatest potential difference across the capacitor?

52. *Why is the following situation impossible?* The LC circuit shown in Figure CQ32.8 has $L = 30.0$ mH and $C = 50.0$ μ F. The capacitor has an initial charge of 200 μ C. The switch is closed, and the circuit undergoes undamped LC oscillations. At periodic instants, the energies stored by the capacitor and the inductor are equal, with each of the two components storing 250 μ J.

53. The switch in Figure P32.53 is connected to position *a* **AMT** for a long time interval. At $t = 0$, the switch is thrown to position *b*. After this time, what are (a) the frequency of oscillation of the LC circuit, (b) the maximum charge that appears on the capacitor, (c) the maximum current in the inductor, and (d) the total energy the circuit possesses at $t = 3.00$ s?

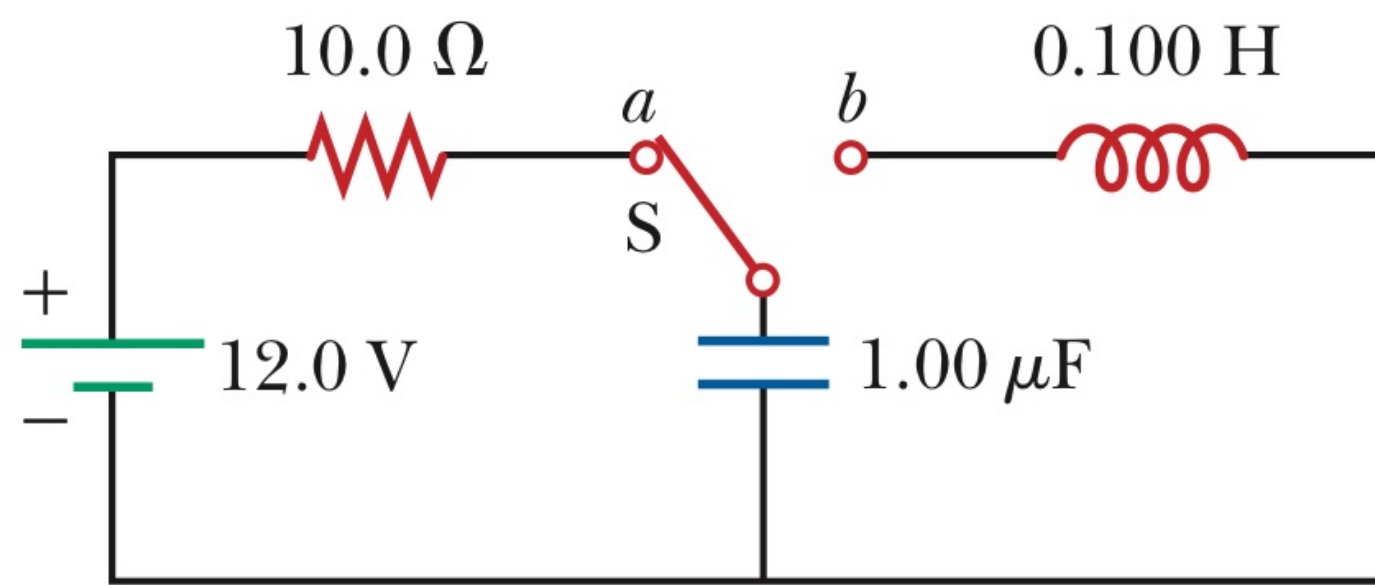


Figure P32.53

- 54.** An LC circuit like that in Figure CQ32.8 consists of a **M** 3.30-H inductor and an 840-pF capacitor that initially carries a $105\text{-}\mu\text{C}$ charge. The switch is open for $t < 0$ and is then thrown closed at $t = 0$. Compute the following quantities at $t = 2.00\text{ ms}$: (a) the energy stored in the capacitor, (b) the energy stored in the inductor, and (c) the total energy in the circuit.

55. An LC circuit like the one in Figure CQ32.8 contains **AMT** an 82.0-mH inductor and a $17.0\text{-}\mu\text{F}$ capacitor that initially carries a $180\text{-}\mu\text{C}$ charge. The switch is open for $t < 0$ and is then thrown closed at $t = 0$. (a) Find the frequency (in hertz) of the resulting oscillations. At $t = 1.00\text{ ms}$, find (b) the charge on the capacitor and (c) the current in the circuit.

56. Show that Equation 32.28 in the text is Kirchhoff's loop rule as applied to the circuit in Figure P32.56 with the switch thrown to position b .

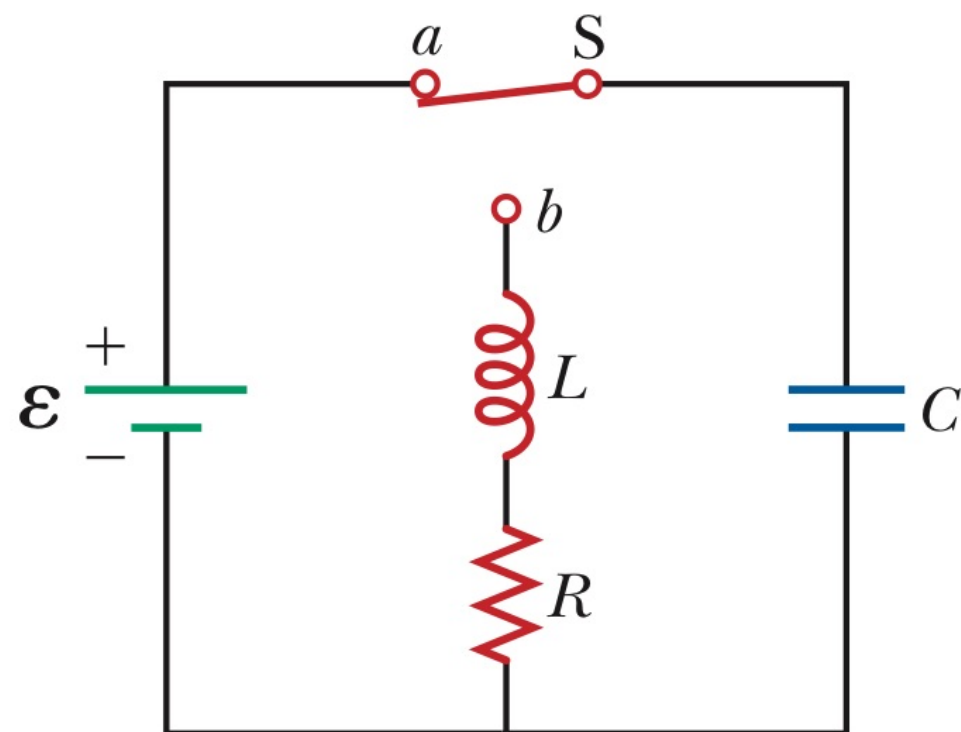


Figure P32.56 Problems 56 and 57.

- 57.** In Figure P32.56, let $R = 7.60\ \Omega$, $L = 2.20\ \text{mH}$, and $C = 1.80\ \mu\text{F}$. (a) Calculate the frequency of the damped oscillation of the circuit when the switch is thrown to position b . (b) What is the critical resistance for damped oscillations?

- 58.** Consider an LC circuit in which $L = 500 \text{ mH}$ and $C = 0.100 \mu\text{F}$. (a) What is the resonance frequency ω_0 ? (b) If a resistance of $1.00 \text{ k}\Omega$ is introduced into this circuit, what is the frequency of the damped oscillations? (c) By what percentage does the frequency of the damped oscillations differ from the resonance frequency?

- 59.** Electrical oscillations are initiated in a series circuit containing a capacitance C , inductance L , and resistance R . (a) If $R \ll \sqrt{4L/C}$ (weak damping), what time interval elapses before the amplitude of the current oscillation falls to 50.0% of its initial value? (b) Over what time interval does the energy decrease to 50.0% of its initial value?

- 63.** A capacitor in a series LC circuit has an initial charge Q and is being discharged. When the charge on the capacitor is $Q/2$, find the flux through each of the N turns in the coil of the inductor in terms of Q , N , L , and C .

68. *Why is the following situation impossible?* You are working on an experiment involving a series circuit consisting of a charged $500\text{-}\mu\text{F}$ capacitor, a 32.0-mH inductor, and a resistor R . You discharge the capacitor through the inductor and resistor and observe the decaying oscillations of the current in the circuit. When the resistance R is $8.00\ \Omega$, the decay in the oscillations is too slow for your experimental design. To make the decay faster, you double the resistance. As a result, you generate decaying oscillations of the current that are perfect for your needs.